william morris

developer && devops in vision and audio

07455205052
william.dunstan.morris@gmail.com
https://williamdunstanmorris.github.io
williamdunstanmorris

current position

Software Developer, Centre for Vision, Speech and Signal Processing, University of Surrey, September 2017 - present
Part of an EPSRC research project (named S3A) with BBC Research & Development in one of the best departments for Computer Vision in the world, in collarboration with the Universities of Southampton and Salford, the project is focused on integral research in object-based audio for immersive home environments. I form part of the development team to engineer cross-platform applications for the audio research and creative communities.

selected projects

Software: Reverberant Spatial Audio Software VST
`C++ & JUCE`
As part of a collaborative effort to produce software for object-based audio, we iteratively integrated our object-based audio rendering framework with a third party C++ framework, JUCE, to produce a package of VST3 softwares audio plugins. The implementation of a VST software allows audio 'objects' to be panned for multiple speaker setups within digital audio workstations. This was achieved through metadata architecture - parsing large JSON format datasets in real-time.

Software: *VISR* **C++/Python Continuous Integration**
`Gitlab-CI & CMake`
To achieve versioned cross-platform releases of an object-based audio framework, called VISR (http://www.s3a-spatialaudio.org/visr), I worked with the University of Southampton remotely to craft an automated pipeline with Gitlab-CI that worked within a successful git branching model, from building in CMake, Unit testing through to packaging. In order to automate the generation of an online API for our C++ framework, we integrated Sphinx and Doxygen with CMake within our stack, and continuously deployed static HTML content on our public web-servers.

Software: *Kluster*
`Max/MSP & OpenGL`
Self-lead project to create a number of abstracted extendable modules that give more user-friendly control over the way audiovisuals are rendered with OpenGL and GLSL in real-time by exploring decoding of MIDI messages. A small but scalable ecosystem was designed to allow routing of midi messages that were decoded and converted into suitable vector or fragement matrices, useful for opengl. I was trying to explore concepts surrounding syncronisity. This project has just lead me to discern more about audio and analytics, and to create a new project, titled Voodoo, which will either be in `Python` or `Golang.`

Software: *Publication Management System*
`Java`
I acquired fundamental concepts of Object-Orientated engineering throughout my MSc, and I built on this through development of softwares such as a Study Planner Application and a Publication Manage- ment System in Java, both collaboratively and individually. I explored different design patterns, like a model-view-controller pattern. As one of my first languages, I apply much of these object-orientated principles in practice with other languages nowadays.

Web: williamdunstanmorris.github.io
`Ruby`
As a way to publish research and regular blogs, I use Github pages and Jekyll to generate a static HTML website. I built this as a way of experimenting and learning about CSS and Javscript. The website is available at https://williamdunstanmorris.github.io

Web: Multipurpose Wordpress Theme
`PHP`
Owner and developer of a creative multi-purpose theme for Wordpress, targeted at giving website designers more front-end controls over their website layouts using the Wordpress dashboard, by embedding three.js and rendering graphics with WebGL. Good implementation of a class-based structure deployed locally at first, enforcing singleton patterns was utilised to seperate logical concerns, and PHP injections were structured to ensure both readability and reusability.

talks and demonstrations

2019, Object Based Binaural Audio - CVSSP 25th Anniversary
Ran my own demo stand for industry leaders in computer vision, speech and signal processing, an exploration into our-project's object-based audio VSTs - the first ever open-source object-based VST eco-system for audio.

2019, Optimising Workflow Practices with Grid-Computing
In another upcoming initiative, I will be explaining gitlab-ci integration with high throughput computing software, HTCondor, for faster workflow approaches in fields involving machine-learning and DNNs.

2018, Continuous Integration Practices
Rising from my project work on developing a CI pipeline for a C++ framework using CMake, I gave a live coding tutorial reflecting on continuous integration, notably on how it could be combined with more computer intensive research. This was an independent initiative that addressed a gap in a trending technology and led to interest from a broad range of participants.

2018, "Git Poetry"
To better my collegues' understanding of version-control techniques, I gave a live git demonstration on good practices for version control for collaborative workflow. As a result, I feel more confident about pushing changes within a collaborative environment. This is summarised in my blog post, https://williamdunstanmorris.github.io//gitflow-techniques.html

2018, "BBC Sounds Amazing", Object-Based Reverberation Demo
At a BBC event in March 2018, I curated a demo of my audio software within a 9.1 speaker setup to members of the public, including an external auditor from Sennheiser.

education

[MSc] Information Technology, University of York, 2016 - 2017
Exposed to a wide-range of computer science disciplines: Advanced Programming Concepts, Software Engineering, Computer Architecture, High-distributed Performance Computing, Software Testing, Database Systems and Information Systems and Organisations. Self-defined my own 6-month research project that used Max/MSP, called Kluster. This is available on Github.

[BA] Music (2:1) , University of Sussex, 2011 - 2014
Introduced to programming with Supercollider, a C-style syntax, and produced algorithmic compositions and my own virtual software instrument as part of the course project. See my website for more on this. Continue to pursue composition and production as a passion.

additional experience

English Teacher, British Institute of Rome, 2016
Acquired a CELTA teaching qualification, built up teaching skills and acquired a modest understanding of spoken Italian.

Activities Manager, Education First, Brighton, 2015
Organised and managed staff and activities leaders, managing budgets and events in venues across Brighton and south-east England.

References

Dr Philip P Jackson, Reader in Machine Audition, University of Surrey, GU27XH
p.jackson@surrey.ac.uk

Professor Adrian Hilton, Director of CVSSP, University of Surrey, GU27XH
a.hilton@surrey.ac.uk